

Final Concept Report

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Contents

Introduction 3
Need 4
Benefits 4
Competition 5
Approach 6
Specifications 6
Design Concepts & Selection Process 7
Consideration of appropriate constraints, standards & regulations 8
Technical Details 10
Prototyping and Testing Plan 11
Sponsor Acceptance testing 12
Business Plan 12
Multi-Disciplinary Collaboration 13
Future Work 13
Images 15
Python Script Prototype 15
Microsoft Authenticator 22
Microsoft Azure Active Directory B2C 23
Microsoft PowerApps 24
Appendix 26

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Introduction

This report outlines the development and implementation of Swipe-2-Type, a multi-factor authentication (MFA) system designed to replace DUO Mobile in educational institutions. The project began with the goal of creating a system like the Department of Defense's CAC (Common Access Card) for use at Wichita State University. However, as the project progressed and the team met with the sponsor, the idea evolved into a replacement for DUO Mobile. Many students at Wichita State University do not have any form of multi-factor authentication being used when they access university resources such as their financials, degree audits and other confidential information portals. With DUO Mobile, our research found that it was quite easy to sign in without the owner's consent because the sign-in does not require the entry of a pin code, but a simple approve or deny message. This was the main motivation for integrating the concept of a CAC card into the login system for Wichita State University. With further development, WSU (Wichita State University) would be able to integrate this system with the help of Microsoft Authenticator, and implement even more features such as location tracking, building access, crowd control in case of an emergency, and also help save on utilities bills since the building access could be used to shut-off unnecessary utilities when a space is unoccupied. The current prototype is in place, and the report details the features and benefits of Swipe-2-Type. Additionally, the report discusses how the system can be maintained with little technical expertise by an institution interested in deploying it for an annual fee. Overall, this report provides a comprehensive overview of Swipe-2-Type and its potential impact on educational institutions' security and authentication processes.

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Need

Our client and sponsor, Professor Joe Jabara, who serves as the director of the Cybersecurity Hub at Wichita State University, approached us with a request to develop a second-factor authenticator using the WSU ID Shockers card. The primary motivation behind this system's development is that out of the twenty customer interviews conducted, twelve had some problems with the current DUO authentication system, which has been deemed inconvenient, unreliable, and insecure. As the student, faculty, and staff populations at Wichita State University continue to grow, there is an increasing need for enhanced security measures on campus to ensure the safety of all individuals. In response to these concerns, we have developed Swipe-2-Type, a reliable and user-friendly multi-factor authentication system that addresses the growing need for enhanced security measures at WSU and other educational institutions, and financial and healthcare industries.

Benefits

Our main benefits will be security, convenience, and affordability for all users. It will provide benefit by requiring the user to prove something they know, which is their phone password and the One-time password (OTP) on top of something they have, which is their mobile phone and their Common Access Card. Our product will provide the users with convenience by having the user simply tap their ID card on the card reader and enter their OTP password that is provided by their authenticator app such as Microsoft Authenticator or Google Authenticator. Our product will be affordable to all users since we are using technology that is already available within Wichita State University through their WSU ID that has a chip in it and card reader that is already available. With the help of this system, it is possible that students might be able to save money, since the cost of the subscription for DUO Mobile can then be removed from the student, since WSU would be

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using Microsoft Authenticator which is complimentary with the existing license that WSU provides to students, staff, and faculty. With the right implementation, WSU could also spend less on utilities which would in turn reduce the load on the fee structure, which is a benefit for all the parties involved.

Competition

Our main competition is DUO authentication, which is what we have, however it is not as convenient or as secure as our solution since if the user loses access to their phone, the attacker has full capability to user's account. With our solution, the attacker will also be required to have the user's ID card, an extra layer of security. Another competitor that we have is the Department of Defense in-house authentication system, they are just as secure as we are, however it is much more expensive to implement and maintain than our solution. With our system's design, we would use resources available within Microsoft Suite. These include but are not limited to Azure Active Directory (B2C (Business 2 Consumer))*, Microsoft Authenticator and other Azure AD Applications. This design would not infringe on the IP rights of DUO Mobile, Microsoft or any other cybersecurity company since Swipe-2-Type is not designing any resource from scratch. In terms of competition, the use of Microsoft Authenticator that is provided by Microsoft, with which WSU has had a secure relation with, would mitigate the other cybersecurity companies since the cost for using the Microsoft AD Suite is included within the existing subscription. The fee for Swipe-2-Type would be considerably less than what DUO currently charges WSU.

**If the project is implemented by WSU, they might use Azure Active Directory instead of Azure Active Directory (B2C) due to the organization size.*

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Approach

Specifications

To meet the client's requirements, the authentication process should have at least two factors and be easy to use. Our proposed design incorporates shocker cards in NFC smartphones, which can significantly reduce the cost of physical shocker cards. From setting up the account on the smartphone application to entering the pin every time the user swipes the card, the process will be straightforward and user-friendly. The application's interface and features will be intuitive and easy to navigate.

On the engineering side, we will utilize the Microsoft Azure platform to build the authentication servers and DNS (Domain Name Service) server that will host the active directory for users' accounts. These two servers will work together with the smartphone application to authenticate users when they try to log in. Depending on their information technology services, they can choose to have in-house servers or utilize the cloud, resulting in a significant cost reduction and improved security.

The existing DUO Mobile authentication system is not open-source and can be costly, especially for a growing institution like Wichita State University. Our in-house solution will not require monthly fees or subscriptions, and the cost will be significantly lower depending on the number of users. The solution's implementation will improve the safety and security of the campus and provide a more convenient and user-friendly authentication process for the staff, faculty, and students.

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Design Concepts & Selection Process

We have identified three different approaches for optimizing the usage of our design concept. The first approach involves equipping WSU computers across all campuses with a card scanner and linking each user account to their unique serial number contained in their shockers ID card's magnetic strip. Users will swipe their card on the scanner to authenticate their login attempt. Upon validation, the authentication server will send a push notification to the user's smartphone via the authenticator application to enter their pin code. If correct, the user will be successfully logged in.

The second approach targets personal devices, where users will enter their credentials the regular way, but will also receive a Swipe-2-Type notification to enter their 4-digit pin code on their smartphones. The third approach involves integrating the WSU shockers card into NFC-enabled smartphones, allowing users to tap their phones into card scanners instead of swiping their physical cards.

After conducting around eighteen interviews either online or in person around the end of October 2022 with students, faculty, and staff, we have decided to move forward with the first design concept. Our decision is based on the following reasons: Firstly, our customer specifically requested the use of the WSU shockers card as part of the second authentication process. Secondly, none of the interviewed students reported having a second factor authentication linked to their accounts, indicating a need for increased security. Thirdly, all interviewed users confirmed they always carry their shockers card with them on campus.

Our interviews also revealed a potential security flaw. A sizable number of users save their credentials on their browser in their personal devices, including those who use Duo Mobile. Moreover, many users leave their phones unlocked and unattended, increasing the risk of

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unauthorized access to their accounts. In our design, even if a user has saved their credentials on their browser, a threat actor cannot access their account without first entering a 4-digit pin code that only the account owner knows.

With the location of university servers on-campus as well as networked computers on-campus, our primary focus was implementing Swipe-2-Type for campus computers. Based on the response from our interviews, the decision was made to implement Swipe-2-Type in phases, enabling users to get accustomed to the system on campus computers, and then implementing the system on personal devices.

Consideration of appropriate constraints, standards & regulations

For the development of the prototype for the design of Swipe-2-Type, Microsoft Azure Active Directory B2C is being used as the database containing secure student information. The front-end was initially designed using Microsoft Power Platform consisting of Microsoft PowerApps, Microsoft Power Automate and other components of the platform. The consideration to use Microsoft Azure Active Directory B2C and Microsoft Platform was made due to Wichita State University providing licenses to these. With a completed application, it would have been significantly cheaper for WSU to implement this system with the current resources already in place.

The Graphic User Interface (GUI) developed through Microsoft Power Platform showed early signs of success due to it being within the Microsoft Environment. The integration between Microsoft Azure Active Directory B2C and Microsoft Power Platform was simple to establish. The goal was to deploy a working application within the Power Platform communicating directly with the Active Directory to a user's smartphone and demonstrate the authentication process.

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During the development process of Swipe-2-Type, the team underwent extensive research to determine the most appropriate front-end for the system. After careful consideration, it was determined that using an existing authenticator such as Microsoft or Google Authenticator would be the most effective approach. The team adapted the process to integrate the chosen authenticator with the system, enabling users to access their accounts securely and conveniently through the established platform. This decision was made after careful consideration of the benefits provided by using an established and widely used authenticator, including the enhanced security features and user convenience. By incorporating this existing technology into Swipe-2-Type, the system can provide users with a reliable and user-friendly multi-factor authentication process. For this purpose, a python script was needed to communicate with a website to enable login using username and password. To set up an account for a user, the information needed to be entered on the Azure AD B2C. Once this was done, the user received an email with a QR Code to be scanned using an authenticator application. Once this was done, the user account was linked to the authenticator on the mobile device. After this, whenever the user tried to login to the website, a username and password would be required. If the username and password entered were correct, a code would be received on the authenticator and the code would need to be entered on the computer. A test for this was run using a local host. Once this test was successful, the team began hosting the website on the cloud and connected it with the Azure AD B2C and the authenticator application.

The user would need to use their WSU ID Card instead of their username and password on their computer. Due to this, a card swipe machine was used to test, enable capturing card information and linking it on the Azure AD B2C to keep users from entering their username and password in public.

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Keeping both the approaches in mind, the decision was made to move forward with the approach using an existing authenticator application, instead of designing a front end on MS PowerApps. The reason for this was the convenience in the application being implemented within Wichita State University or other institutions. Users would be more likely to look forward to using already established authenticators by companies like Google and Microsoft. With their immense security already in place, these authenticators would work perfectly for this project, while being able to demonstrate the security provided by Swipe-2-Type.

Technical Details

Our design consists of an interplay between the front-end smartphone authentication app and the back-end MFA server, in such a way that the encrypted communications between them allows secure access to the WSU system.

Originally, we started working on developing our own authenticator, but due to time limitations, and with the approval of our client, Prof. Jabara and our mentors, Dr. Huabo and Dr. Brookings, we decided to drop the application development and focus all of our time and energy on the core systems, that is the MFA server and the windows active directory on Azure. And instead, we can use any available authentication app in the market, like Google or Microsoft authenticator. In addition, the front-end space also consists of the webpages that users will need to access their WSU accounts.

The active directory business to consumer (AD B2C) hosts our domain on Azure, under the name spmfa.onmicrosoft.com, referring to Senior Project MFA. Within this AD, we have created a database of users who can use their AD accounts to authenticate. User accounts on AD can save a lot of valuable information that can be used when users try to log in, so in this case, and in addition

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to the username and password, each AD account will also save the serial number of the corresponding WSU ID card. In a typical manner, WSU webpage uses AD accounts to authenticate users by linking the webpage to the AD, and when users click on the login button, they are redirected to that specific AD, and depending on the working environment, they will either swipe their WSU ID cards or type in their username and password. Next to the Active directory, we also created a linked MFA server that will be used to further authenticate users before allowing them to log in. This MFA server is written in Python and uses "pyotp" library to generate and verify One-Time Passcode (OTP). When users enable MFA, a secret file will be released and linked to the user account, along with an associated QR code that can be scanned and automatically added to the authenticator app. This will result in a six digits code that will be changed every 30 seconds. From this point forward, every successful login attempt will always consist of a combination of valid credentials and the OTP.

Prototyping and Testing Plan

For testing purposes, the login webpage, user accounts and the MFA server are all created locally. Deploying this system on Azure is especially important to make it accessible from anywhere in the world. To do so, we need to create an app registration within our AD on azure, then we can use this app registration to configure the webpage and the MFA server.

Currently, we are in the process of finishing these last few steps, but to test the app, we can navigate to the app directory in CMD and type: *python manage.py runserver* to run the MFA server than we can navigate to the localhost webpage where users can log in using their usernames and passwords. Once logged in, they can navigate to the profile tab where they can enable MFA and

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scan the QR code on their authenticator. From that point forward, users will need to their OTP along with their usernames and password to login.

Sponsor Acceptance testing

At this point, we are ready to present our customer with a working prototype that satisfies the criteria of Professor Joe Jabara. The full demo that we presented, included the following:

First, we created a user account, then we tested the user account on the login webpage. We used the username “Joe” and his WSU ID card as the password. We logged in, and enabled MFA, then using an authenticator on his phone, he scanned the QR code to link his account to his authenticator. Then he tested the log in process again. He typed his username, swiped his WSU ID for password, and he used the 6 digits pin from his authenticator to log in.

Professor Jabara was very satisfied with this working prototype and wanted to find a way to further fine tune it, so it would be an acceptable project so that WSU or other institutions might consider deploying it for students, faculty and staff. He also wanted to see if next year's students in the senior design class can work on it some more and present a different version of it.

In the Engineering Open House Day, he visited our station multiple times, and he was very interested in knowing what other people are saying about our project and suggested writing a paper for a technical document for a paper such as the IEEE.

Business Plan

Swipe-2-Type is designed to be a system with minimal third-party work. This means that an institution would be able to design and maintain their database internally. Depending on the size of the organization, as well as the current security protocols of the organization, deploying

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Swipe-2-Type would involve purchasing encrypted cards for each member, purchasing card readers to read the encrypted cards and setting up the Active Directory. The cost of setting up the Active Directory would vary depending on whether it is implemented on the cloud on on-premises. Our team would perform the initial set-up and guide the designated business personnel on how to modify and maintain the system, as needed. Due to the nature of this system, Swipe-2-Type would be based on a monthly or annual subscription plan, based on the needs of the organization. Our team would be available for support in the future in case an issue occurs after the system has been deployed.

Multi-Disciplinary Collaboration

Computer science and cybersecurity are working closely together in this project, with cybersecurity engineering a more secure system and computer science implementing a secure code and turning this design into a functional product. For computer science, we got both a partial application and a fully working prototype website. To be able to get both applications done, we had Hemil do an amazing job of getting frameworks that we can use a skeleton to base our application. The computer science influence can be seen with the website's design in a user-friendly approach. On the cybersecurity part, the focus was linking the user account and the login system to user's ID card that encodes a large amount of information and linking all of this system to an authentication app on user's smartphone to make it more secure.

Future Work

Our project can be continued if it is adopted by Wichita State University or any institute. Our project has been pretty much complete with a demo which only needs to be implemented and adapted to any institute. Our project can be used with any popular authentication application that

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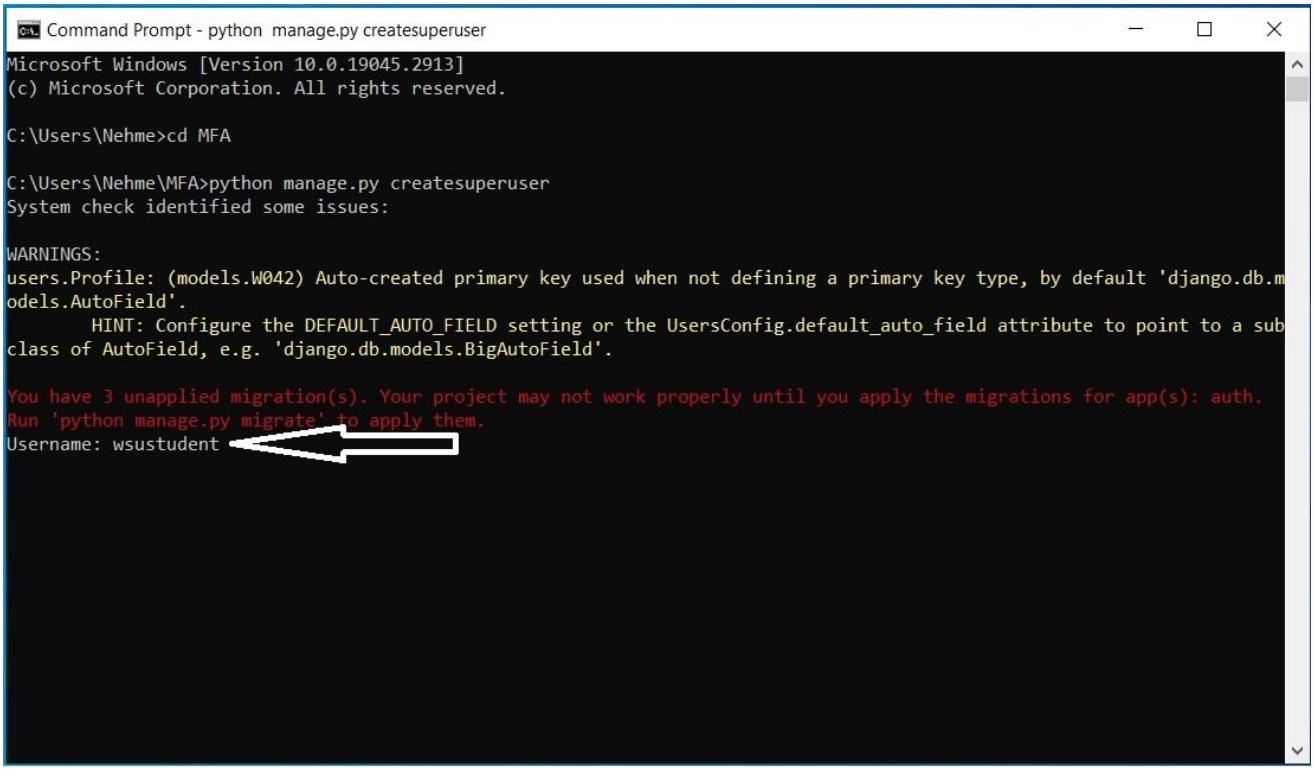
is available on the App Store or Google Store, so every user will have access and will be able to use our secure system. In the future, to deploy our system, the institution can link their website or application after authenticating their users through the Active Directory, which would then redirect them to our system. In addition, institutions will also need to publish common access cards for their own employees and purchase card readers to implement and maximize the security capability that our project has.

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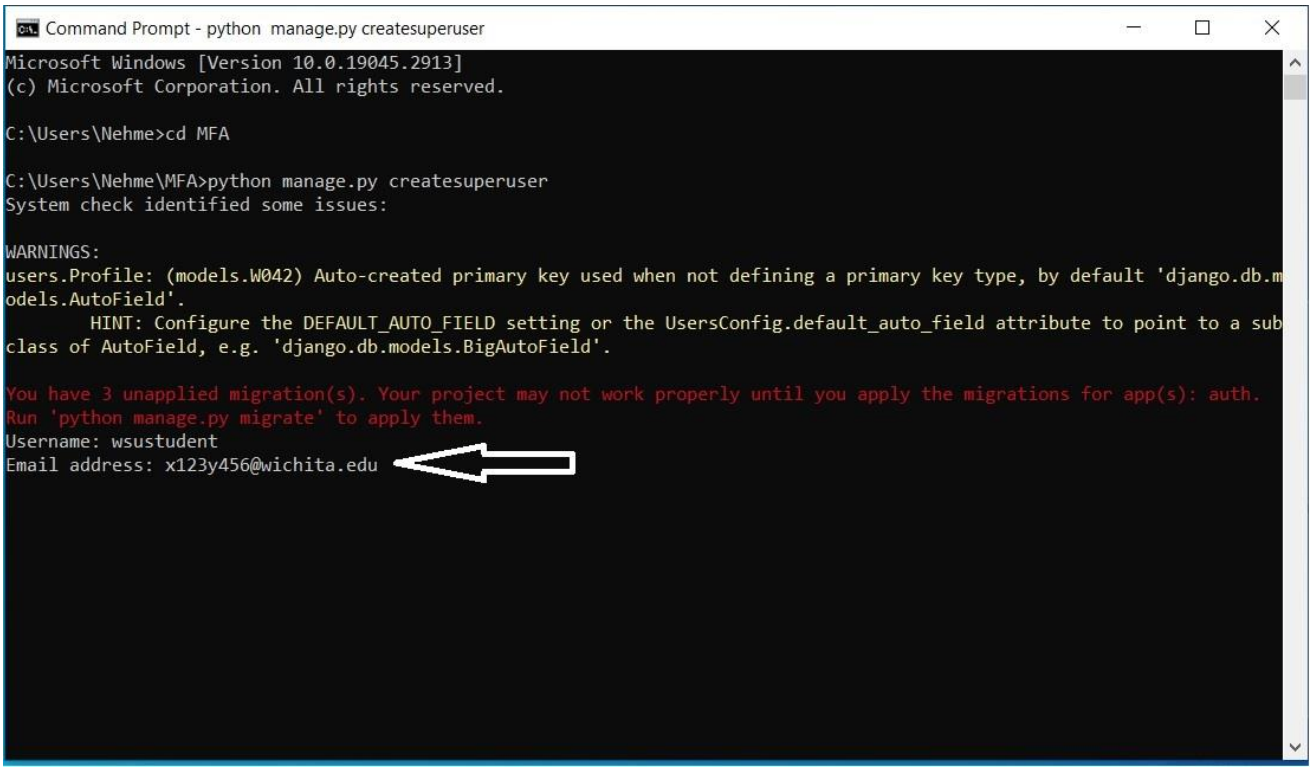
Images

Python Script Prototype

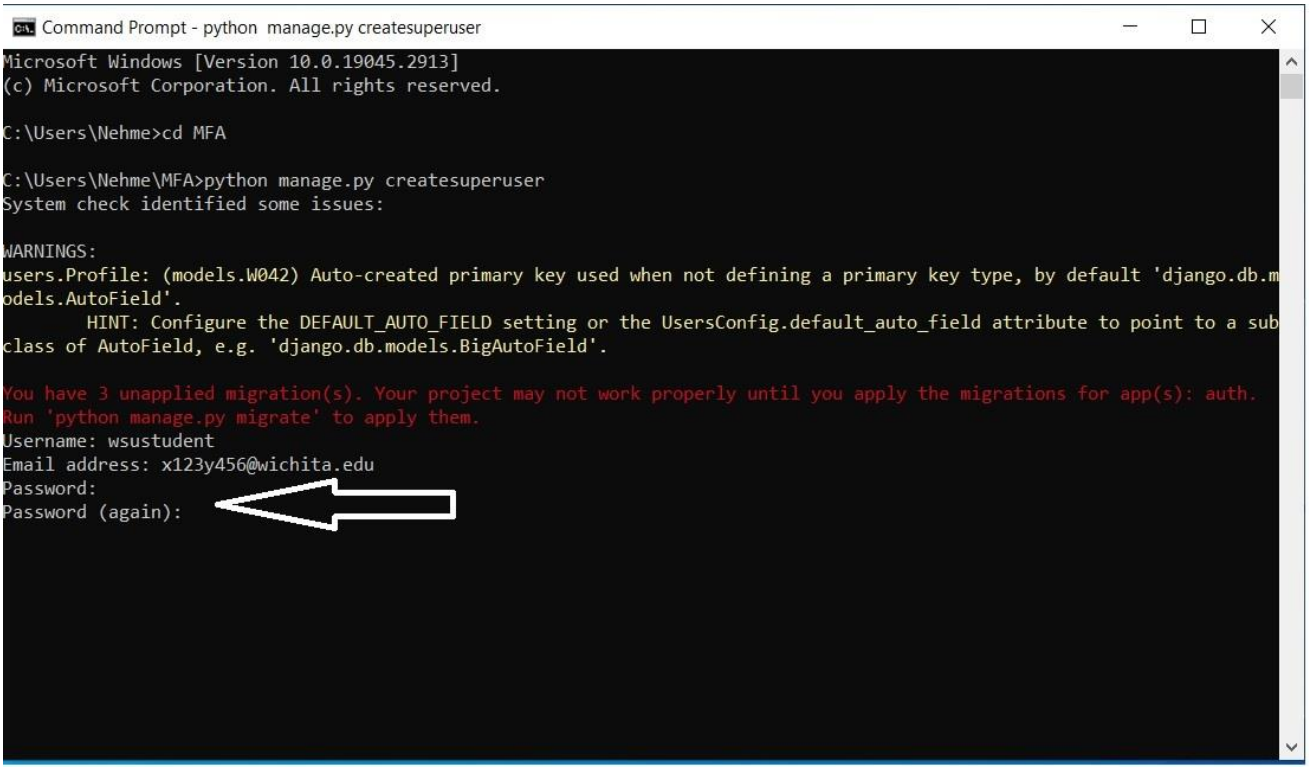
1- We create a user “wsustudent”



2- We enter student’s email address.

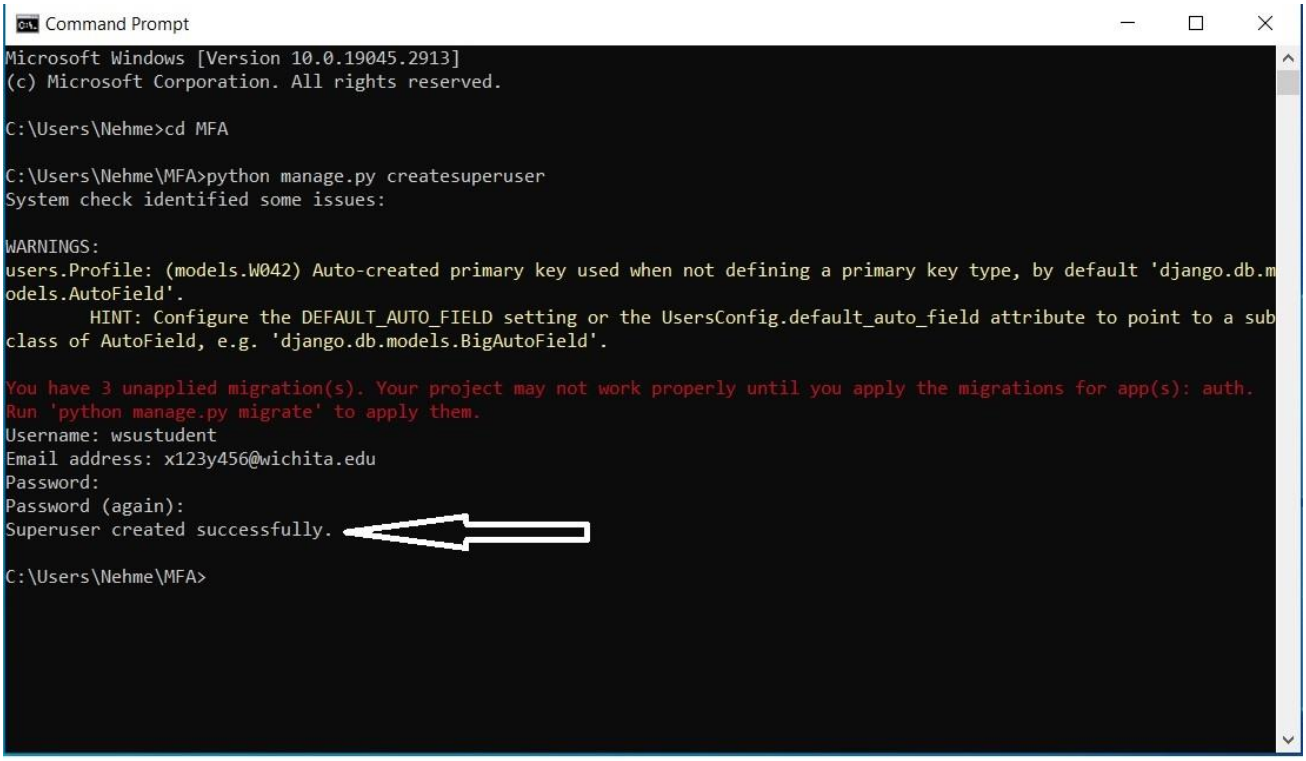


3- We swipe our WSU ID card for the password (Twice to confirm password)

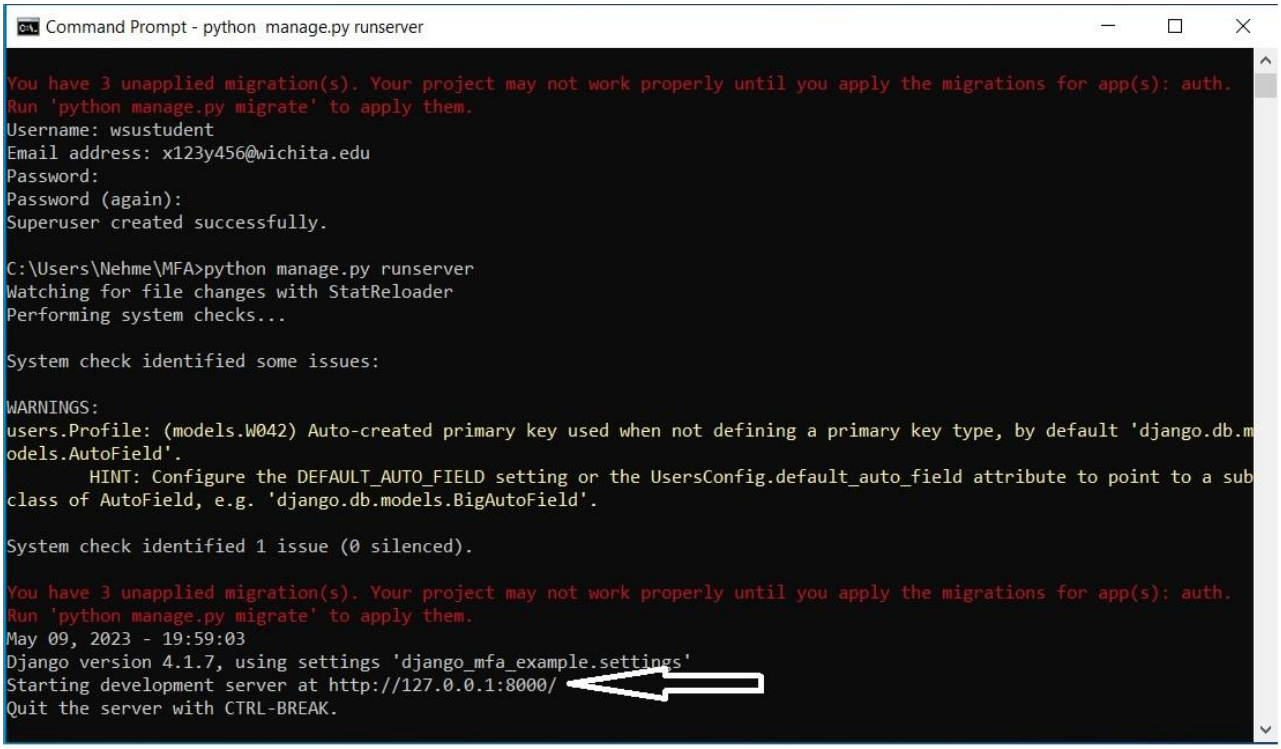


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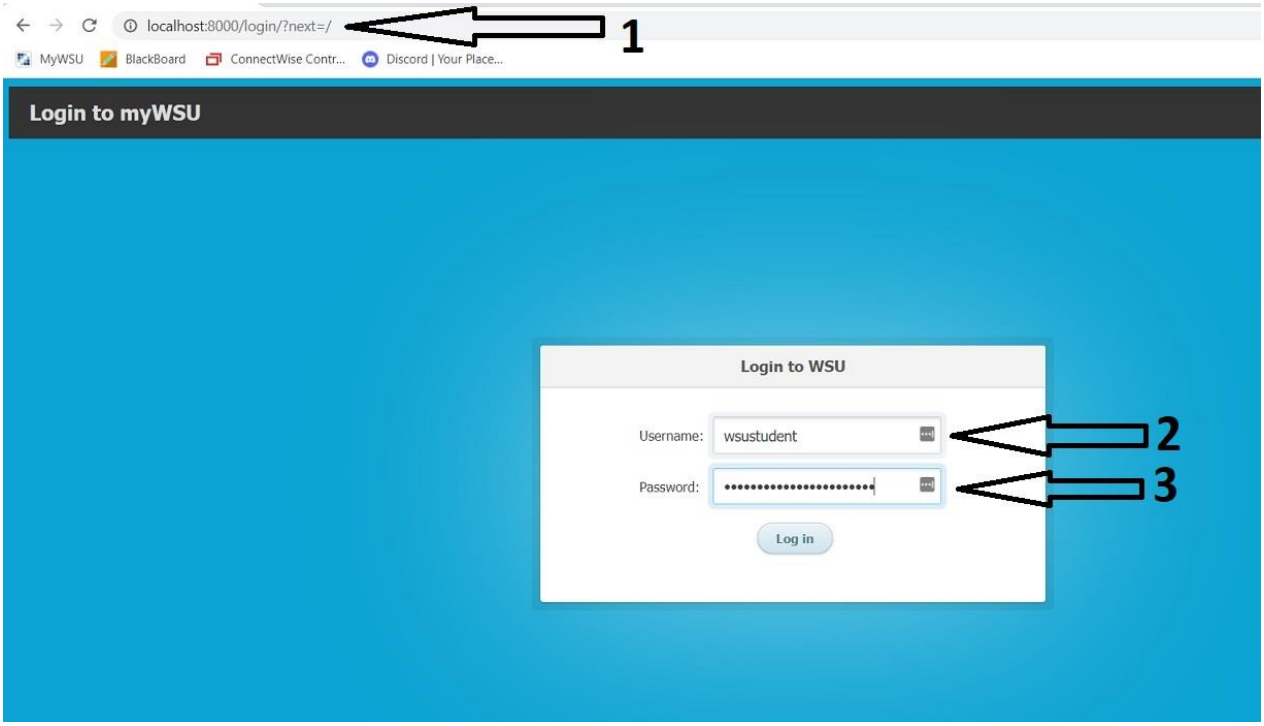
3- User account created successfully.



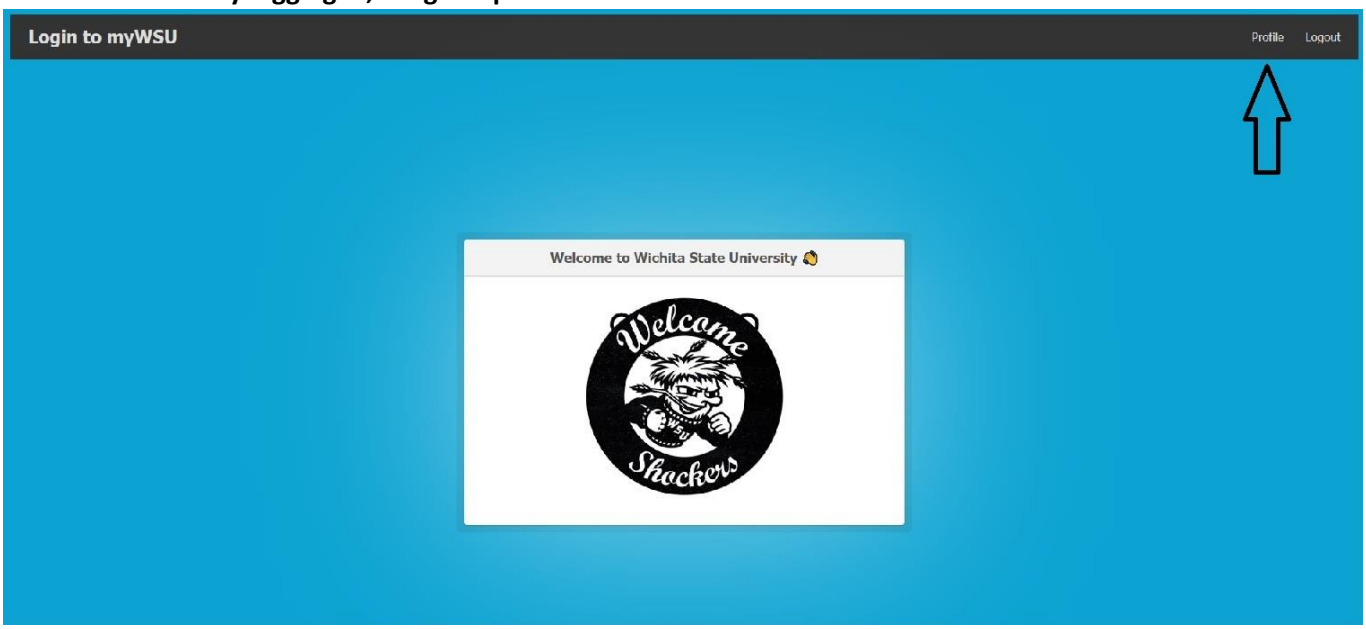
4- Now we run the server using the local address.



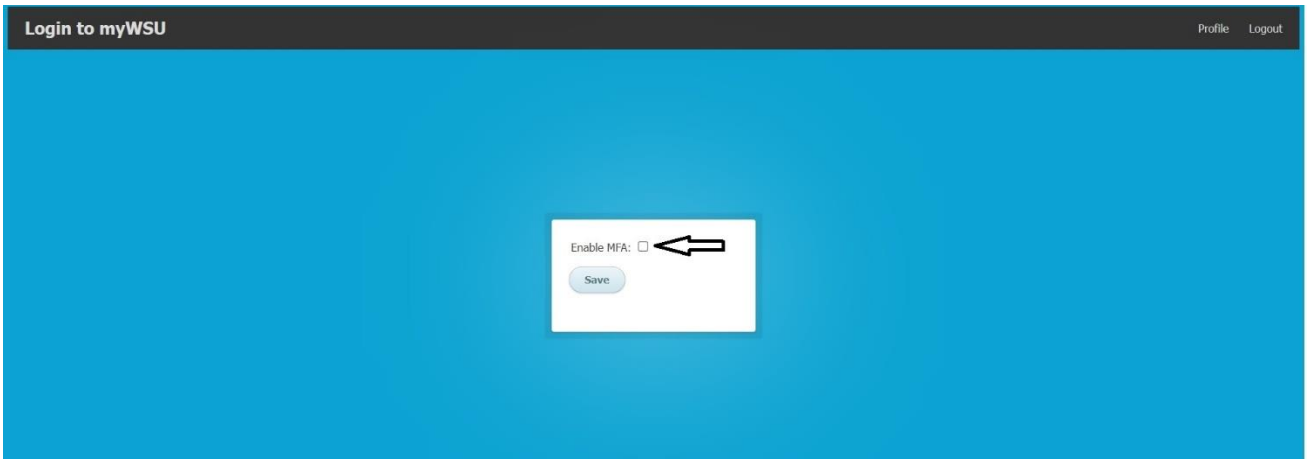
5- Now, on the browser we go to the local webpage and we login using the username and swiping our WSU ID card for password



6- after successfully logging in, we go to profile tab to enable MFA



7- we check the Enable MFA box and the click on "Save" to generate the QR code

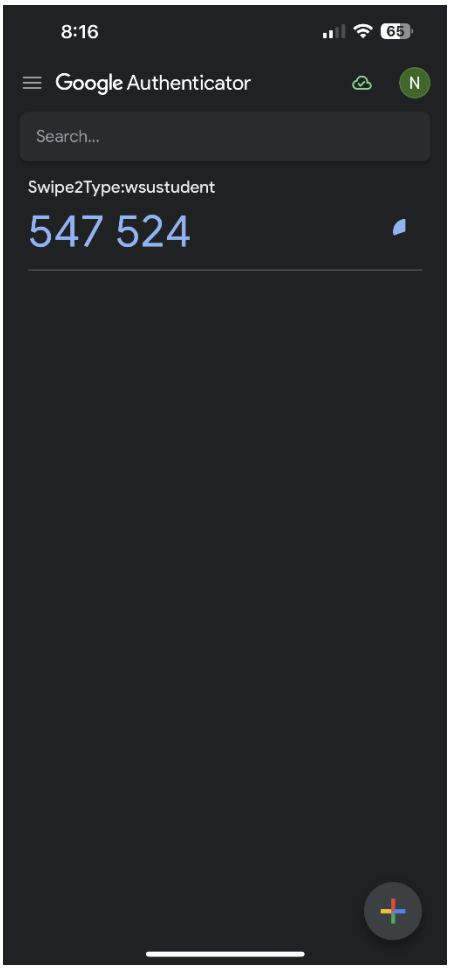


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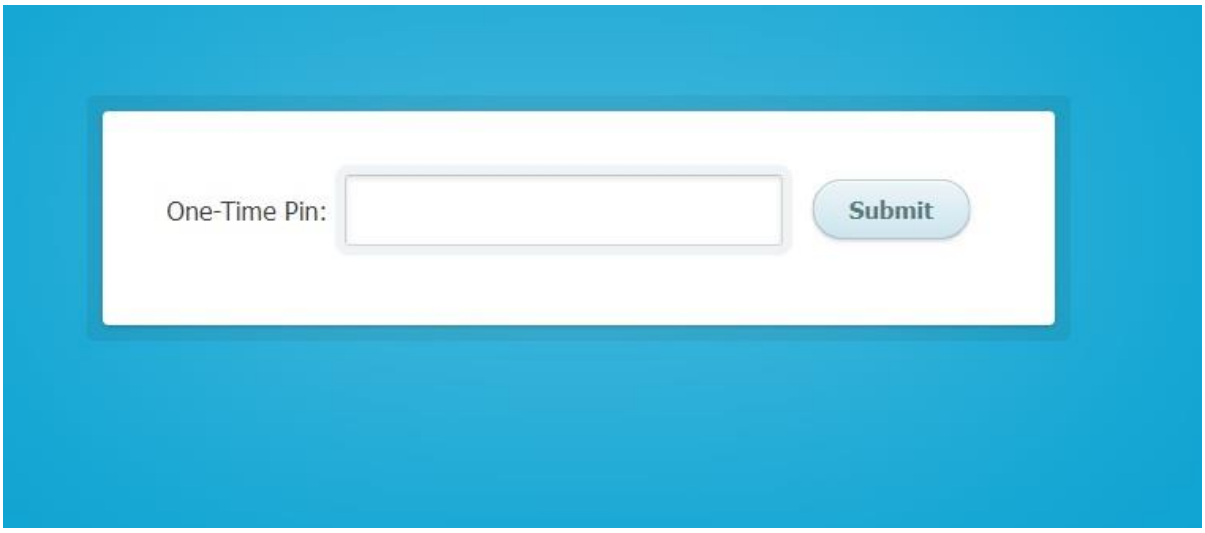


8- We can scan this QR code using any authentication app (here I used Google authentication).

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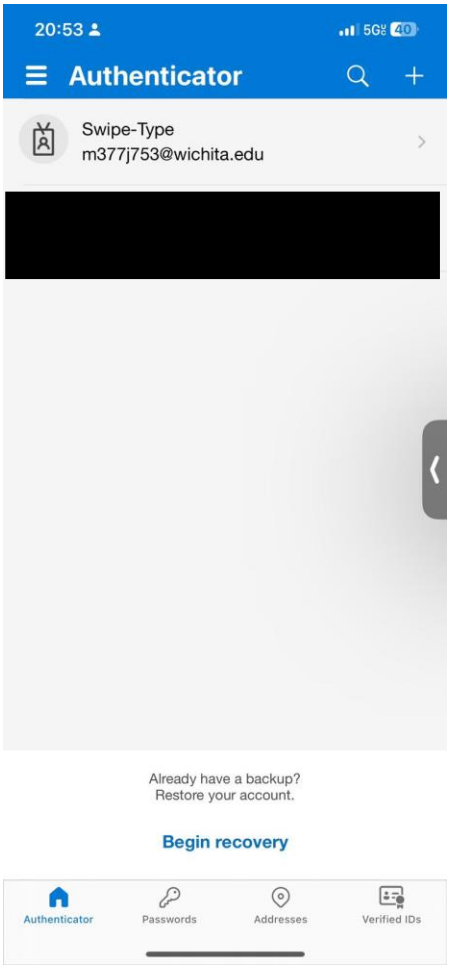


9- Now, wsustudent account is linked to the smartphone. The second time we need to log back in, we need to type in the username, swipe the WSU ID card for the password, and we also need to enter the 6-digits pin from the authenticator.



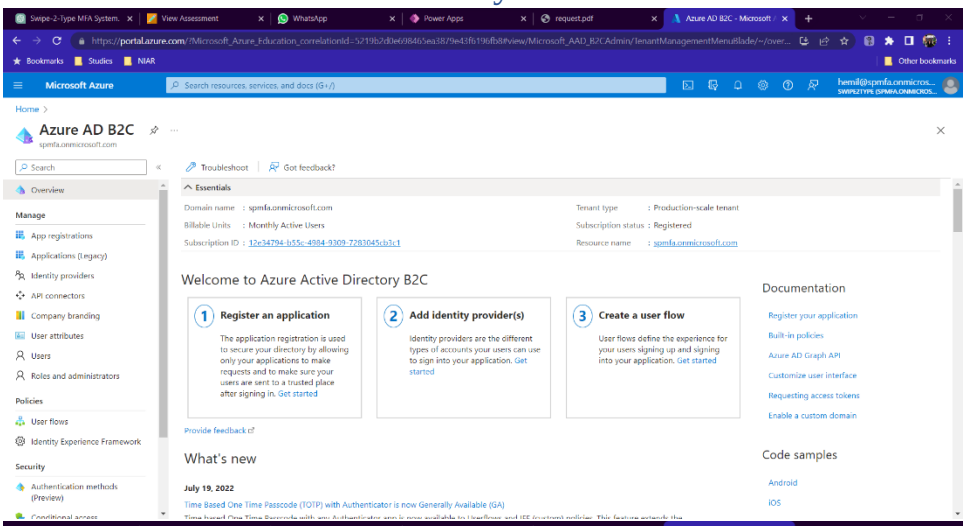
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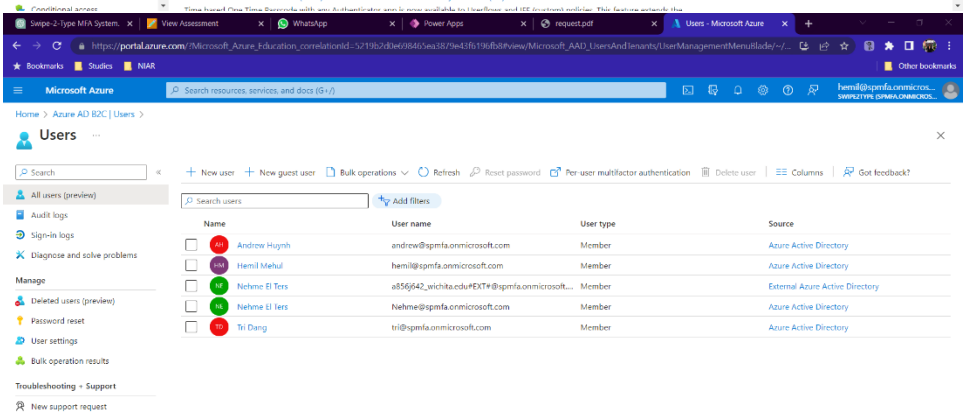
Microsoft Authenticator



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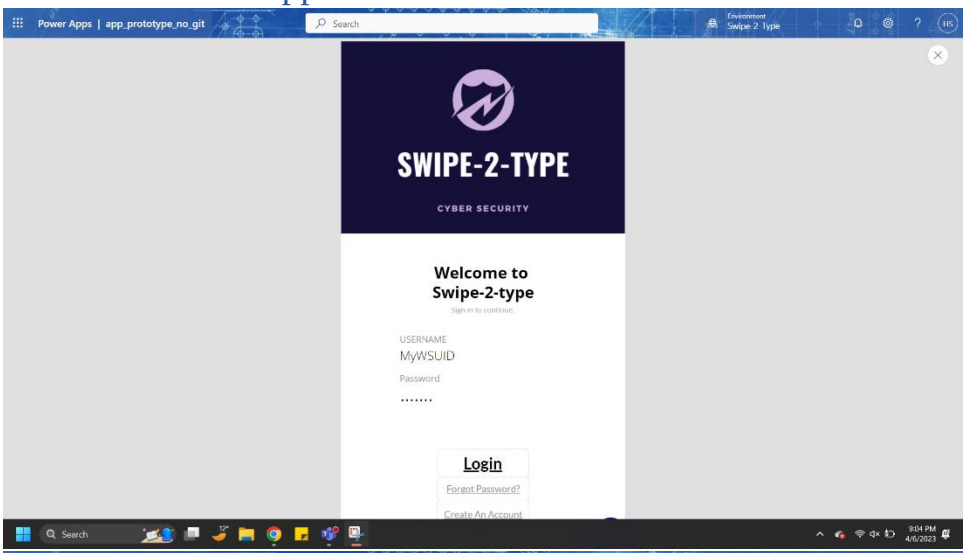
Microsoft Azure Active Directory B2C

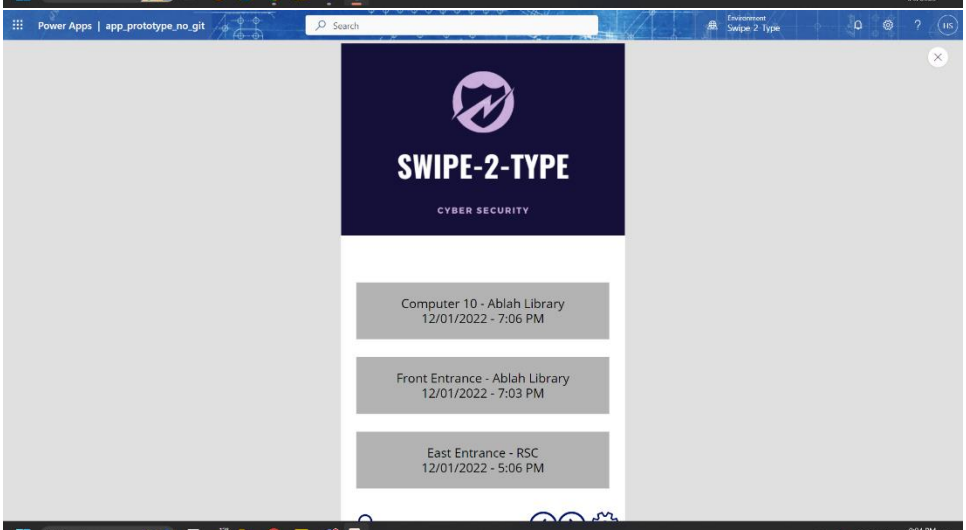


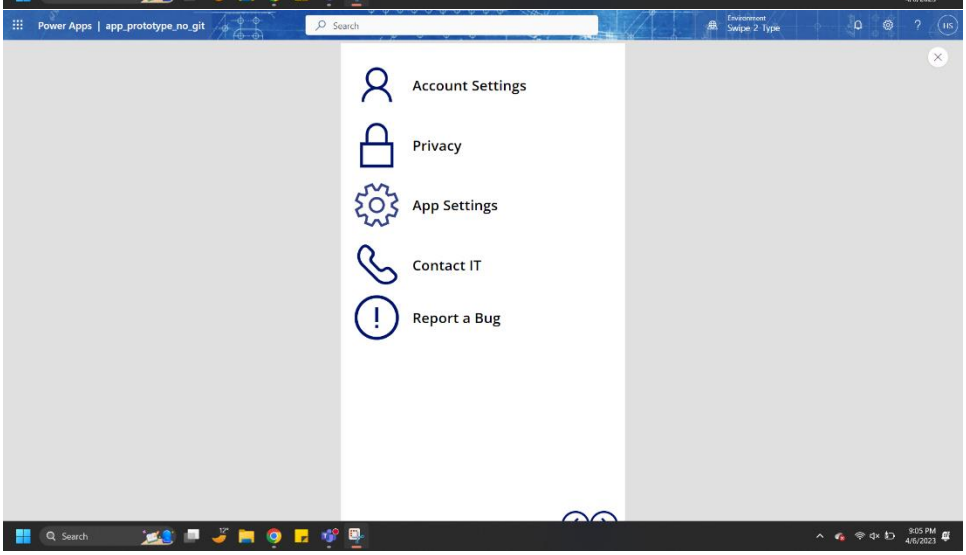


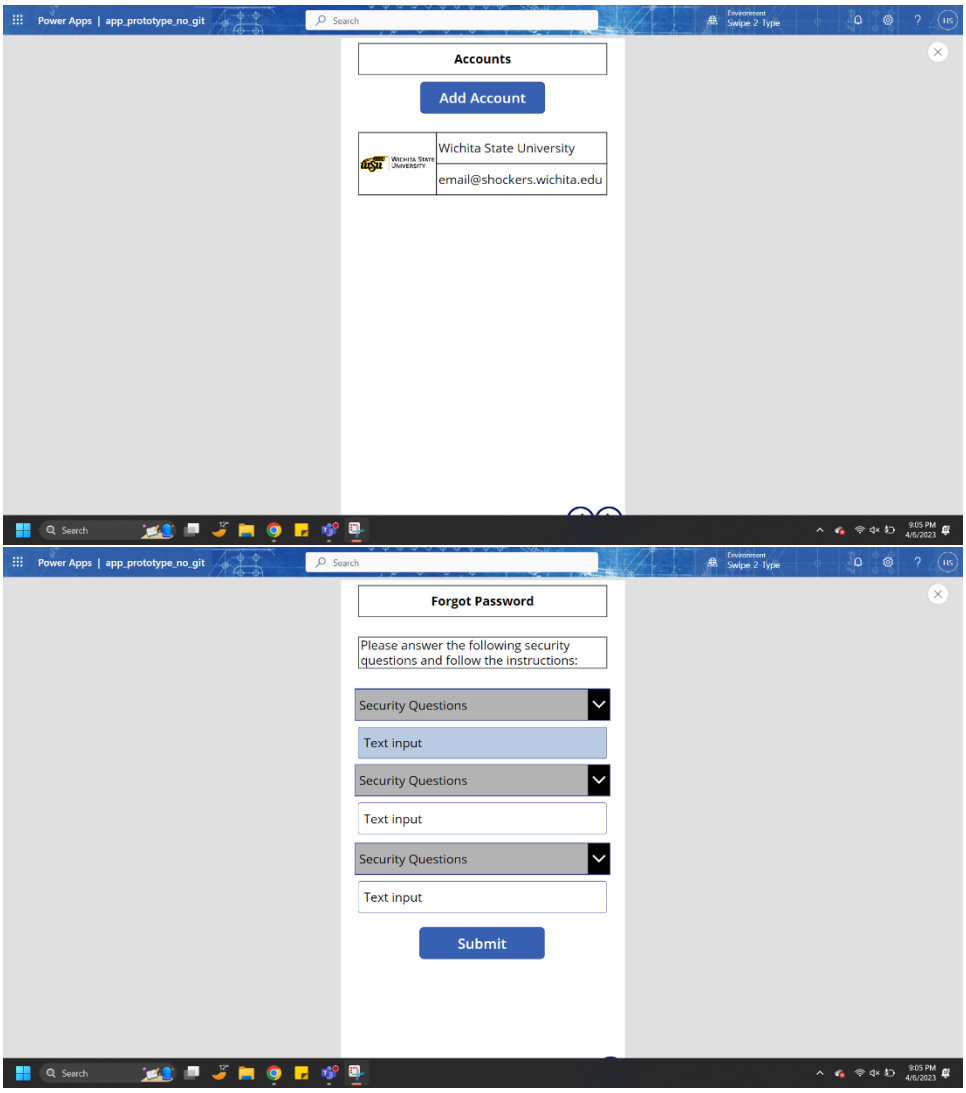
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Microsoft PowerApps








Power Apps | app_prototype_no_git
Search
Environment
Swipe 2 Type
Accounts
Add Account
Wichita State University
email@shockers.wichita.edu
Forgot Password
Please answer the following security questions and follow the instructions:
Security Questions
Text input
Security Questions
Text input
Security Questions
Text input
Submit

Appendix

The team consisted of three students from the Applied Computing department and one student from the Computer Science department. Overall, this division provided the team with opportunities to cover the bases and the overall needs of the system.

The tasks were divided as follows:

Hemil Mehul Shah & Andrew Huynh: Develop Microsoft PowerApps GUI and assist with back-end as needed.

Nehme El Ters & Tri Dang: Building the Active Directory in Azure Business to Consumer (B2C) and integrate with a webpage, integrate the database with an existing authenticator.

Entire team: Develop the software and corresponding python code.

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